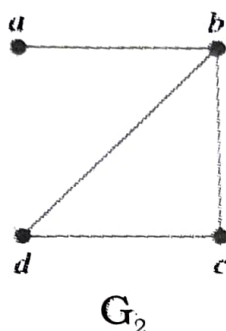
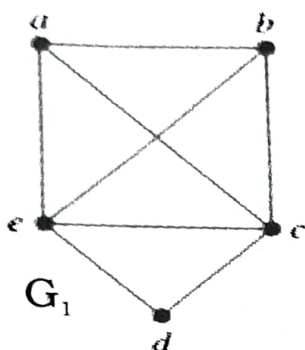


Section - A

1. (a) Define a Hamilton path and Hamilton circuit. List a Hamilton path or Hamilton circuit in the graphs G_1 and G_2 given below if it exists.

5



- (b) (i) Find the contrapositive, the converse and the inverse of the statement :
"The home team wins whenever it is raining". 3
- (ii) Show that if n is an integer and $3n + 2$ is odd, then n is odd using a proof by contraposition. 2
- (c) Find the greatest common divisor of 414 and 662 using the Euclidean algorithm. 5
- (d) Let f be the function from the set $X = \{2, 3, 4, 5, 6, 7\}$ into the set $Y = \{0, 1, 2, 3, 4\}$ defined by $f(x) = 2x \pmod{5}$. Write f as a set of ordered pairs. Is f one - one or onto ? Justify. 5

- (e) (i) How many distinct 5 – letter passwords can be formed from English alphabets if repetition is allowed and at least one letter must be a vowel ? 3
- (ii) How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points ? 2
- (f) Let 'a' be a numeric function such that : 5

$$a_r = \begin{cases} 1 & 0 \leq r \leq 3 \\ 2^r + 3 & r \geq 4 \end{cases}$$

Find ∇a and $S^{-2} a$.

Section - B

2. (a) In how many ways can three examinations be scheduled in a seven-day week if more than one examination can be scheduled on the same day. 3
- (b) Let R be a relation on the set of real numbers such that 4

$$R = \{(a, b) \mid a^2 + b^2 = 1\}$$

State whether the relation R is reflexive, symmetric, antisymmetric and transitive. Briefly justify your answer in each case.

- (c) Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3 for every integer $n \geq 1$. 8
3. (a) Let $N(x)$ be the statement "X has visited North Dakota", where the domain consists of the students in your school. Express each of these quantifications in English. 3
- (i) $\exists x N(x)$
- (ii) $\forall x N(x)$
- (b) Find the sum and the product of $(1001)_2$ and $(1101)_2$ without using decimal number system. 4
- (c) In a survey of 60 people, it was found that 25 people read Hindi newspaper, 26 read English newspaper, 26 read Urdu newspaper, 9 read both Hindi and Urdu newspaper, 11 read both Hindi and English newspaper, 8 read both English and Urdu newspaper, 3 read all three newspapers. Find : 8
- (i) The number of people who read at least one of the three newspapers.
- (ii) The number of people who read exactly one newspaper.
- (iii) The number of people who read both Hindi and Urdu newspaper, but not English ?

4. (a) (i) Define planar graph. Is K_5 a planar graph? Justify your answer. 5

(ii) Suppose that a connected planar graph has 30 edges. If a planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have? 12 2

(b) Let 'a' and 'b' be two numeric functions defined as follows : 8

$$a_r = \begin{cases} 2 & 0 \leq r \leq 2 \\ 2^{-r} + 5 & r \geq 3 \end{cases}$$

$$b_r = \begin{cases} 3 - 2^{-r} & 0 \leq r \leq 1 \\ r + 2 & r \geq 2 \end{cases}$$

Determine the sum and product of numeric functions a and b.

5. (a) In how many ways can a committee of 5 members be formed from 11 Senators so that : 7

(i) Senator A is always included

(ii) Senator A is always excluded

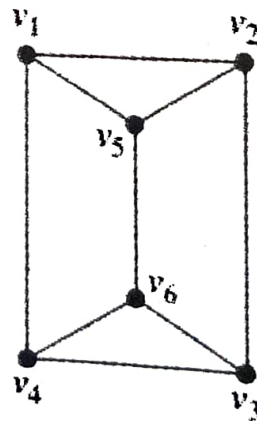
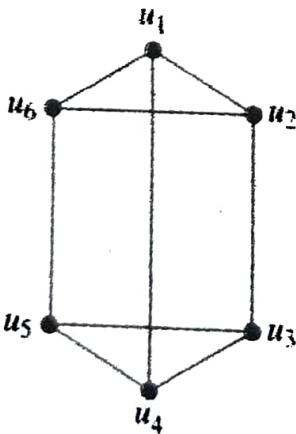
(iii) At least one of senator A and senator B will be included

- (b) Consider the following premises : 8

^A "It is not sunny this afternoon and it is colder than ^B yesterday", "We will go swimming only if it is sunny", "If we do not go swimming, then we will take a canoe trip", and "If we take a canoe trip, then we will be home by sunset".

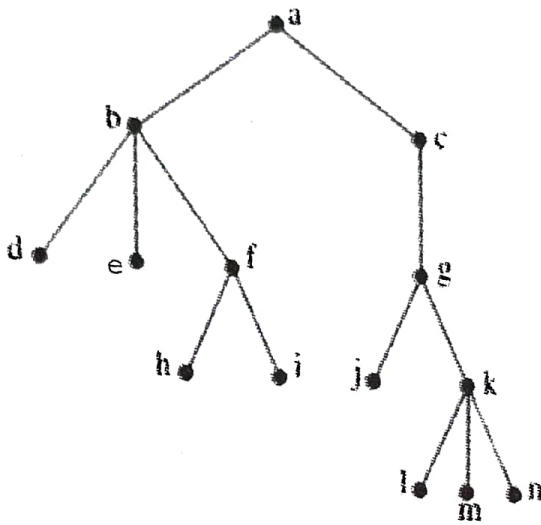
Determine whether the conclusion ^E "We will be home by sunset." follows logically from the above set of premises.

6. (a) What do you mean by graph invariant ? Is the following pair of graphs isomorphic ? Justify. 7



- (b) Evaluate $3^{644} \bmod 645$ using Fast Modular Exponentiation algorithm. 8

7. (a) How many leaves does a full 4-ary tree with 101 vertices have? Given the rooted m-ary tree below, answer the following questions : 7



- (i) Draw the subtree that is rooted at g ?
 - (ii) Which vertices are ancestors of k ? a, c,
 - (iii) What is the value of m for the given rooted m -ary tree? 3
 - (iv) Is the tree a balanced tree? Give reason for your answer.
- (b) Given a relation R on set $A = \{3, 5, 9, 15, 24, 45\}$ such that : 8
- $$R = \{(a, b) : a \text{ is divisor of } b \text{ and } a, b \in A\}$$
- (i) Show that R is a POSET.

5796

- (ii) Draw Hasse Diagram.
- (iii) Find the maximal and minimal elements in A.
